

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Technology (B.Tech)

Semester – I / II (First Year) University Examination May 2013

B.Tech. (CE/EC/IT) / B.Tech. (CL/EE/ME)

CE101 Fundamentals of Computing & Programming (FOCP)

Date: 04/05/2013, Saturday

Time: 01:30 p.m to 4:30 p.m

Maximum Marks:70

## Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.

## SECTION – I

**Q.1(a) State the major difference between following with suitable example with respect to 'C' Programming Language (Any Three).** 06

1. Variables and Constant.
2. Arrays and Structures.
3. Actual arguments and Formal arguments.
4. Recursion and Iteration
5. Compiler and Interpreter

**Q.1(b) What is the correct output of the following code?**

04

```
(1) union student{ int rollno; int item;
    };
void main(){
    union student s;
    s.rollno=8; s.item=9;
    s.item=10;
    printf("%d %d %d",s.rollno, s.item, s.item);
}

(2) void main()
{
    int q=3;
    switch(q)
    {
        case 1: printf("I");
        case 2: printf("LIKE");
        case 3: printf("C");
        case 4: printf("AND");
        default: printf("C++");
    }
}
```

**Q.1(c) What is the difference between the statements *rewind(fp)* and *fseek(fp, 0L, 0)*?**

01

**Q.2 Answer the following questions.**

12

1. Explain in brief about entry controlled loop and exit controlled loop with an example.
2. Narrate the usage of pointer. Also define scale factor.
3. Pen down the significance of *strlen()*, *strcmp()*, *strcpy()*, *strcat()* functions.

OR

**Q.2 Answer the following questions.**

12

1. Explain: Various types of operators in C Programming language.
2. Draw a flow chart to exchange the value of two variables without using third variable.
3. Explain the meaning and purpose of: *tag*, *struct*, *local variable*, *global variable*



- Q.3 Write programs in C.** 12
1. Write a C program Write a program that takes contents of a file and copied them into another file and print it on the screen.
  2. Write a program to find out given year, which is leap year or not. A leap year is year which is evenly divisible by 4, but if it is evenly divisible by 100 then it is not a leap year, but if it is evenly divisible by 400 then it is a leap year.

OR

- Q.3 Write programs in C.** 12
1. Write a program to check whether the given number is perfect or not using do...while loop.
  2. Write a program that will display the contents of a file and count how many characters, spaces, tabs and new lines are present in it.

## SECTION – II

- Q.4 Do as directed.** 11
1. Replace the following code using if...else statements. 03  
`ans = (a1>a2) ? ((a1>a3)? a1 : a3 ) : a2;`
  2. Explain the use of sizeof. 02
  3. Convert the given decimal number to binary number: 123 02
  4. When a program is terminated, all the files used by it are automatically closed. 02  
 Why is then necessary to close a file during execution of program?
  5. (i) Any string in C ends with \_\_\_ symbol which is known as \_\_\_ character. 02  
 (ii) What is the difference between two operators = and ==?

- Q.5 Answer the following questions.** 12
1. Explain the basic structure and steps of execution of a C program.
  2. Explain: Two dimensional array with an example

OR

- Q.5 Answer the following questions.** 12
1. Explain: Types of functions in C programming language
  2. Explain: Various types of looping in C programming language

- Q.6 Write programs in C.** 12
1. Write a function exchange to interchange the values of two variables without using pointer.
  2. Define a structure that describes a hotel. It should have member that included name, address, grade, average room charge, and number of rooms. Write a function to print out all hotels of a given grade in order of charges. Also print out hotels with room charges less than given values.

OR

- Q.6 Write programs in C.** 12
1. Write a function program to sum of odd numbers between 1 to n, where n is entered through keyboard.
  2. Define a structure called cricket that will describe: player name, team average and batting average. Using cricket, declare an array player with 50 elements and write a program to read the information about all the 50 players and print team-wise list containing names of players with their batting average.

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